HOTEL

HOTEL BUSINESS ANALYTICS

INVESTIGATING HOTEL BUSINESS USING DATA VISUALIZATION

Executive Business Intelligence Report



A comprehensive analysis of hotel booking patterns, revenue performance, customer behavior, cancellations and seasonality to generate actionable business insights and strategic recommendations.

119,390+	$7.12M	2	234,988+	2017 – 2019
Total Bookings (Before Cleaning)	Realized Revenue	Hotel Types (City & Resort)	Total Guests	Analysis Period

TOOLS & TECHNOLOGIES

Pandas · Python · NumPy · Matplotlib · Seaborn · Streamlit

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Project
Hotel Business Analytics

“Turning data into insights, insights into actions, and actions into better business.”

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1. Executive Summary

This project presents a data-driven analysis of hotel booking behaviour to understand demand patterns, seasonality, customer behaviour and cancellation risk. The analysis uses a historical hotel booking dataset covering **2017–2019** and applies Python-based data analytics techniques using Pandas, NumPy, Matplotlib and Seaborn, with Streamlit used for presenting analytical outputs.

The analysis begins with data-quality assessment and preprocessing, including duplicate detection, missing-value treatment, categorical-value standardisation and validation of anomalous records. Business-focused features such as total stay nights, total guests, stay-duration categories and booking lead-time categories were developed to support deeper analysis.

The results show that **City Hotel is the dominant booking type**, accounting for approximately two-thirds of the bookings in the original dashboard view. Overall cancellation behaviour is a major business concern, with the supplied dashboard reporting a **37.0% cancellation rate** across 119,390 bookings. City Hotel also demonstrates a higher cancellation rate than Resort Hotel. Seasonality is another important factor, with **October identified as the peak booking month** in the dashboard analysis.

One of the strongest findings is the relationship between **booking lead time and cancellation risk**. The dashboard shows cancellation increasing from approximately **9.6% for bookings made 0–7 days in advance to 57.0% for bookings made more than 180 days in advance**. This indicates that far-ahead reservations represent a particularly important risk segment.

Overall, the analysis demonstrates how hotel booking data can support better revenue planning, inventory management and cancellation control. The key recommendation is to implement a **lead-time-based cancellation-risk strategy**, supported by proactive confirmation, flexible rescheduling, appropriate cancellation policies and seasonal pricing. These actions can improve occupancy predictability, reduce cancellation-related uncertainty and strengthen overall hotel business performance.

Key Performance Indicators

KPI	Value	Business Significance
Total Bookings	**119,390**	Overall booking volume
Cancellation Rate	**37.0%**	Indicates significant cancellation exposure
City Hotel Share	**66.4%**	Dominant hotel booking segment
Average Lead Time	**104 days**	Indicates substantial advance-booking behaviour
Peak Booking Month	**October**	Highest seasonal demand
Highest Lead-Time Cancellation	**57.0%**	Critical high-risk booking segment

2. Introduction

The hospitality industry operates in a highly competitive and demand-sensitive environment where effective decision-making depends on understanding customer booking behaviour. Hotels must continuously manage room availability, pricing, occupancy, cancellations and customer demand while responding to seasonal fluctuations and changing booking patterns. In this context, historical booking data provides an important source of information for identifying operational risks and improving business performance.

This project, **Hotel Booking & Cancellation Behaviour Analysis**, focuses on analysing hotel reservation data from **2017 to 2019** to identify meaningful patterns in booking demand and cancellation behaviour. The analysis covers two major hotel categories, **City Hotel and Resort Hotel**, and examines important booking characteristics such as arrival month, lead time, length of stay, customer type, deposit type, market segment and cancellation status.

The project follows a structured data analytics methodology beginning with data inspection and quality assessment. The dataset is examined for duplicate records, missing values, inconsistent categorical information and unusual values. Appropriate preprocessing and feature engineering techniques are then applied to prepare the data for business analysis. Key derived variables, including total stay nights, total guests, stay-duration categories and lead-time categories, are used to make the analysis more meaningful and easier to interpret.

The main purpose of the project is to answer three important business questions: **which hotel type receives the highest booking demand, whether length of stay influences cancellation behaviour, and whether booking lead time affects cancellation risk**. In addition, the project investigates seasonal demand and customer behaviour to provide broader business context.

The analysis is implemented using **Python, Pandas, NumPy, Matplotlib and Seaborn**, with **Streamlit** used to present the analytical results through an interactive dashboard. The final objective is to transform historical booking data into actionable insights that can support better **revenue planning, inventory management, cancellation control, pricing decisions and customer retention strategies**.

3. Business Problem & Objectives

3.1 Business Problem

The hotel industry operates in an environment where demand, customer behaviour and booking decisions can change significantly across properties and seasons. One of the major operational challenges is managing **booking cancellations**, as unexpected cancellations can reduce occupancy, create inventory uncertainty and affect revenue forecasting. At the same time, hotels need to understand which customer and booking characteristics are associated with higher cancellation risk.

The project addresses this challenge using historical hotel reservation data from **2017–2019**, covering both **City Hotel and Resort Hotel**. The analysis focuses on booking demand, seasonal patterns, length of stay and booking lead time to identify measurable factors associated with cancellation behaviour.

The central business problem is therefore:

How can historical hotel booking data be used to understand demand patterns and identify booking characteristics associated with cancellation risk, so that hotels can make better operational and revenue-management decisions?

The analysis also considers differences between hotel types and examines how booking characteristics can support more targeted cancellation-management strategies.

3.2 Project Objectives

The primary objective is to transform raw hotel booking data into meaningful business insights that can support management decision-making. The project specifically aims to:

1. Identify which hotel type receives the highest booking demand.
2. Analyse monthly booking patterns and identify seasonal demand trends.
3. Determine whether **length of stay** is associated with cancellation behaviour.
4. Determine whether **booking lead time** is associated with cancellation risk.
5. Compare cancellation behaviour between City Hotel and Resort Hotel.
6. Identify high-risk booking segments requiring proactive management.
7. Develop practical recommendations for cancellation control, pricing, inventory planning and customer management.
8. Present the findings through analytical visualisations and an interactive **Streamlit dashboard**.

Business Objective Framework

Objective Area	Key Measure	Business Purpose
Booking Demand	Hotel booking share	Identify dominant hotel type
Seasonality	Monthly bookings	Support demand and capacity planning
Cancellation	Cancellation rate	Measure booking-risk exposure
Stay Behaviour	Total stay nights	Assess relationship with cancellation
Lead Time	Lead-time category	Identify high-risk advance bookings
Business Action	Recommendations	Improve revenue and operational decisions

4. Key Business Questions

The analysis is designed around a set of business questions that connect hotel booking data with practical management decisions. The questions focus primarily on **booking demand, cancellation behaviour, stay duration and booking lead time**. Answering these questions helps identify where the hotel business faces demand opportunities and cancellation-related risks.

The project addresses three core questions specified in the project brief. First, it examines **which hotel type customers book most often**, comparing City Hotel and Resort Hotel to understand the overall booking mix. Second, it investigates **whether length of stay affects cancellation rate**, using total stay nights to compare short, medium and long-stay bookings. Third, it evaluates **whether booking lead time affects cancellation rate**, with bookings grouped into meaningful advance-booking categories.

Additional supporting questions are used to provide business context, including identifying seasonal booking patterns, comparing cancellation rates between hotel types, understanding customer and booking characteristics, and identifying high-risk booking segments.

The answers to these questions are intended to move the project beyond descriptive reporting. They provide a basis for recommendations related to **cancellation management, demand planning, pricing, inventory control and customer retention**.

Key Business Questions Framework

No.	Business Question	Analytical Focus	Business Decision
1	Which hotel type is booked most often?	City vs Resort booking share	Demand & resource planning
2	Does length of stay affect cancellation?	Stay duration vs cancellation rate	Cancellation control
3	Does lead time affect cancellation?	Lead-time groups vs cancellation rate	Booking-risk management
4	When is demand highest?	Monthly booking trends	Seasonal planning
5	Which segments have higher risk?	Customer/channel characteristics	Targeted intervention

5. Dataset Overview

The **Hotel Booking Dataset** used in this project contains historical reservation information for two hotel categories: **City Hotel** and **Resort Hotel**. The dataset covers bookings made for arrivals between **2017 and 2019** and provides detailed information about booking characteristics, customer behaviour, stay duration, pricing and cancellation status.

The dataset contains **119,390 records and 29 variables** in its original form. These variables provide a broad view of hotel reservation activity, including booking lead time, arrival dates, number of guests, length of stay, average daily rate (ADR), deposit type, market segment, customer type and whether a booking was cancelled.

For this project, the dataset is particularly useful because it combines both **demand-related and behavioural variables**. This allows the analysis to move beyond simple booking counts and investigate the factors associated with cancellation risk. The `hotel` variable is used to compare City Hotel and Resort Hotel, while `is_canceled` serves as the primary outcome variable for cancellation analysis. The `lead_time` and stay-night variables are used to investigate booking timing and length-of-stay behaviour.

Before analysis, the dataset was subjected to data-quality checks covering duplicate records, missing values, inconsistent categorical values and anomalous observations. Additional analytical variables were created during preprocessing to improve business interpretation.

Dataset Profile

Dataset Attribute	Description
Dataset Name	Hotel Booking Dataset
Original Records	**119,390**
Original Variables	**29**
Analysis Period	**2017–2019**
Hotel Types	City Hotel, Resort Hotel
Primary Outcome	Booking Cancellation
Data Type	Historical Hotel Reservation Data

Key Analytical Variables

Variable	Business Meaning	Analytical Use
`hotel`	Hotel category	Hotel comparison
`is_canceled`	Cancellation indicator	Cancellation analysis
`lead_time`	Days between booking and arrival	Cancellation-risk analysis
`arrival_date_month`	Arrival month	Seasonality analysis
`stays_in_weekend_nights`	Weekend stay nights	Stay-duration analysis
`stays_in_weekdays_nights`	Weekday stay nights	Stay-duration analysis
`adr`	Average Daily Rate	Pricing analysis
`market_segment`	Booking market segment	Channel analysis
`customer_type`	Customer classification	Customer behaviour
`deposit_type`	Deposit arrangement	Cancellation analysis

6. Data Quality Assessment

Before performing exploratory data analysis, the hotel booking dataset was assessed to identify potential data-quality issues that could affect the reliability of business insights. The assessment focused on **duplicate records, missing values, inconsistent categorical values, invalid numerical values and unusual booking observations**.

The original dataset contained **119,390 records and 29 variables**. A major issue identified during the assessment was the presence of **33,261 exact duplicate records**. Duplicate observations can artificially increase booking volume and distort cancellation rates, therefore they required careful treatment before the main analysis.

Missing values were also present in several variables. The `company` column contained a particularly large proportion of missing observations, while `agent`, `city` and `children` also contained missing values. These missing values do not necessarily represent errors; in many cases they indicate that a particular attribute was not recorded or was not applicable to the booking.

Categorical consistency was another consideration. The `meal` column contained an `Undefined` category, which required interpretation before analysis. Numerical validation also identified **negative ADR values**, which are not meaningful for normal hotel pricing analysis. In addition, records with zero total guests were reviewed because they do not represent a conventional guest booking.

The assessment therefore established a controlled data-preparation process rather than deleting information without justification. Each issue was reviewed according to its potential business impact, and appropriate treatment was applied during the subsequent preprocessing stage.

Data Quality Findings

Data Quality Issue	Finding	Potential Impact
Duplicate Records	**33,261**	Inflated booking and cancellation metrics
Missing `company`	**112,593**	Incomplete company information
Missing `agent`	**16,340**	Incomplete agent information
Missing `city`	**488**	Incomplete location analysis
Missing `children`	**4**	Minor guest-count issue
Undefined Meal	Present	Ambiguous category interpretation
Negative ADR	**1 record**	Invalid pricing value
Zero-Guest Records	**180 records**	Potentially invalid booking observations

Data Quality Assessment Summary

Quality Dimension	Assessment	Priority
Completeness	Missing values in selected fields	Medium
Uniqueness	Large number of duplicate records	**High**
Consistency	Undefined categorical values	Medium
Validity	Negative ADR and zero-guest observations	High
Reliability	Requires preprocessing before analysis	**High**

Overall, the assessment confirms that **data cleaning is essential before drawing business conclusions**. The identified issues were documented and addressed systematically in the Data Cleaning & Preprocessing stage.

7. Data Cleaning & Preprocessing

Data cleaning and preprocessing were performed to convert the raw hotel booking dataset into a reliable dataset suitable for business analysis. The process followed the issues identified during the data-quality assessment and focused on improving data consistency without unnecessarily removing useful information.

First, **33,261 exact duplicate records** were identified and removed because duplicate bookings could inflate booking volume, cancellation rates and other calculated metrics. Missing values were then handled according to the meaning of each variable. Missing values in `children` were replaced with **0**, assuming no children were recorded for those bookings. Missing `city` values were labelled as **"Unknown"**, while missing `agent` and `company` values were represented as **"No Agent"** and **"No Company"** respectively. These treatments preserve the observations while clearly identifying unavailable information.

The `meal` variable contained an `Undefined` category. This was recategorised as **"No Meal Information"** to make the category more understandable during visualisation and interpretation.

Numerical validation was also performed. The dataset contained **one negative ADR record**, which was removed because a negative average daily room rate is not meaningful for normal hotel pricing analysis. Records with **zero total guests** were also excluded from the main behavioural analysis because they do not represent a conventional guest booking.

Finally, new analytical variables were created, including **Total Stay Nights**, **Total Guests**, **Stay Category** and **Lead-Time Category**. These derived fields made it easier to compare cancellation behaviour across different booking patterns.

Cleaning & Preprocessing Actions

Issue / Variable	Treatment Applied	Purpose
Duplicate records	Removed 33,261 duplicates	Prevent metric inflation
`children`	Missing values → 0	Complete guest calculation
`city`	Missing → "Unknown"	Preserve records
`agent`	Missing → "No Agent"	Identify unavailable agent data
`company`	Missing → "No Company"	Preserve booking information
`meal`	`Undefined` → "No Meal Information"	Improve category clarity
Negative ADR	Removed 1 record	Ensure valid pricing data
Zero-guest records	Excluded from analysis	Improve behavioural validity

Feature Engineering

Derived Variable	Calculation / Definition	Analytical Purpose
Total Stay Nights	Weekend nights + weekday nights	Measure length of stay
Total Guests	Adults + children + babies	Measure booking size
Stay Category	Short / Medium / Long	Compare cancellation behaviour
Lead-Time Category	0–7 / 8–30 / 31–90 / 91–180 / 181+ days	Identify cancellation-risk groups

The resulting dataset provides a cleaner and more consistent foundation for exploratory data analysis and business decision-making.

8. Feature Engineering

Feature engineering was performed to transform the cleaned hotel booking data into meaningful variables that could support business analysis and visualisation. The objective was not to create unnecessary variables, but to convert existing fields into measures that are easier to interpret from a hotel-management perspective.

The first major feature created was **Total Stay Nights**, calculated by combining weekday and weekend stay nights. This variable provides a single measure of the total length of a customer's planned stay and is used to investigate whether longer bookings have different cancellation behaviour.

A second feature, **Total Guests**, was created by combining adults, children and babies. This provides a more complete representation of booking size and helps identify valid guest reservations.

To make cancellation analysis easier to interpret, Total Stay Nights was converted into three **Stay Categories**: Short Stay (0–2 nights), Medium Stay (3–5 nights) and Long Stay (6+ nights). These categories allow management to compare cancellation behaviour across different types of stays.

The `lead_time` variable was also transformed into five **Lead-Time Categories**: 0–7 days, 8–30 days, 31–90 days, 91–180 days and 181+ days. This segmentation is particularly important because it allows cancellation risk to be evaluated according to how far in advance a reservation was made.

Additional categorical standardisation was retained from the preprocessing stage, including readable labels for missing agent, company and city information. These engineered variables provide a structured foundation for exploratory data analysis and help connect technical analysis with practical business questions.

Engineered Features

Feature	Definition	Business Purpose
Total Stay Nights	Weekend nights + weekday nights	Analyse length of stay
Total Guests	Adults + children + babies	Measure booking size
Stay Category	Short: 0–2, Medium: 3–5, Long: 6+	Compare cancellation by stay length
Lead-Time Category	0–7, 8–30, 31–90, 91–180, 181+ days	Identify cancellation-risk segments
Guest Booking Status	Based on total guest count	Support valid-booking analysis

The resulting features make the dataset more suitable for **descriptive statistics, comparative analysis and business visualisation**, particularly for answering the project's core questions regarding hotel demand, stay duration and booking lead time.

9. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the underlying patterns, distributions and relationships within the cleaned hotel booking dataset before developing the final business insights. The analysis combined descriptive statistics, categorical comparisons and trend-based visualisations to examine booking demand, hotel type, seasonality, customer behaviour and cancellation patterns.

The first area of analysis focused on the **distribution of bookings between City Hotel and Resort Hotel**. City Hotel accounts for the larger share of bookings, indicating stronger overall booking volume in the dataset. This comparison provides an important foundation for understanding differences in demand and cancellation behaviour between the two properties.

The analysis then examined **monthly booking trends** to identify seasonal fluctuations. Booking volumes vary across arrival months, demonstrating that hotel demand is not evenly distributed throughout the year. Identifying peak and low-demand periods can support better capacity planning, pricing decisions and promotional strategies.

A major focus of the EDA was **cancellation behaviour**. Cancellation rates were compared across hotel types, stay-duration categories and booking lead-time groups. The results indicate that cancellation behaviour differs considerably across these dimensions. In particular, bookings made far in advance show substantially greater cancellation exposure than bookings made close to the arrival date.

The relationship between **length of stay and cancellation** was also investigated using the engineered Stay Category variable. Short, medium and long stays were compared separately for City Hotel and Resort Hotel to determine whether longer reservations represent greater cancellation risk.

EDA Areas

Analysis Area	Key Variable(s)	Business Purpose
Hotel Booking Mix	`hotel`	Compare City vs Resort demand
Monthly Demand	`arrival_date_month`	Identify seasonality
Cancellation Behaviour	`is_canceled`	Measure cancellation exposure
Stay Duration	Total Stay Nights / Stay Category	Examine stay-related risk
Lead Time	`lead_time` / Lead-Time Category	Identify high-risk bookings
Customer Behaviour	`customer_type`, `is_repeated_guest`	Understand customer patterns

Analytical Measures

Measure	Purpose
Booking Count	Measure demand volume
Booking Share (%)	Compare hotel/property contribution
Cancellation Rate (%)	Measure booking risk
Average Lead Time	Understand booking behaviour
Average ADR	Assess pricing characteristics
Total Stay Nights	Measure accommodation demand

Overall, the EDA provides the analytical foundation for the subsequent business analysis. The identified patterns are used to answer the project's core questions and develop evidence-based recommendations for **cancellation management, seasonal planning, pricing and inventory control**.

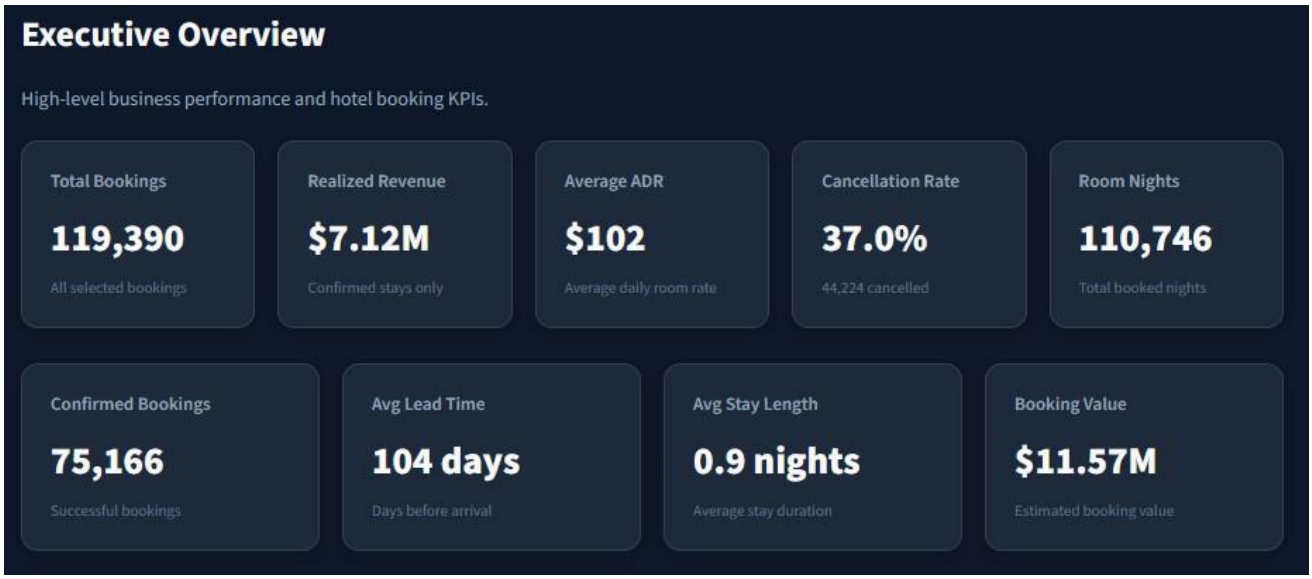


Figure 1: EDA overview

10·Hotel Booking Demand Analysis

Hotel booking demand was analysed to understand the overall distribution of reservations between **City Hotel and Resort Hotel** and to identify patterns in booking volume across the analysis period. Understanding the composition and timing of demand is important for hotel management because it supports decisions related to room inventory, staffing, pricing and promotional planning.

The analysis shows that **City Hotel receives the larger share of bookings**, representing approximately **66.4% of the total bookings** in the supplied dashboard analysis, while Resort Hotel accounts for approximately **33.6%**. This indicates that City Hotel is the primary source of booking volume within the dataset. However, higher booking volume should not automatically be interpreted as higher profitability, since cancellation behaviour, ADR and customer mix also influence business performance.

Monthly booking analysis further shows that demand varies throughout the year rather than remaining constant. **October is identified as the peak booking month**, recording approximately **13,877 bookings** in the supplied dashboard analysis. This seasonal variation indicates that hotel demand is influenced by time of year and should therefore be incorporated into operational and commercial planning.

The difference between hotel types and monthly demand patterns provides useful information for management. During high-demand periods, hotels can focus on improving revenue through appropriate pricing, inventory controls and reduced dependence on unnecessary discounts. During lower-demand periods, targeted promotions and packages can be used to stimulate bookings.

Hotel Booking Demand Summary

Demand Indicator	Result	Business Interpretation
Total Bookings	**119,390**	Overall booking volume
City Hotel Share	**66.4%**	Dominant booking segment
Resort Hotel Share	**33.6%**	Smaller but significant demand segment
Peak Booking Month	**October**	Highest seasonal demand
October Bookings	**13,877**	Strong peak-period booking volume

Overall, the demand analysis demonstrates that **hotel type and seasonality are important dimensions of booking behaviour**. These findings provide a foundation for subsequent analysis of cancellation risk, pricing and customer behaviour.

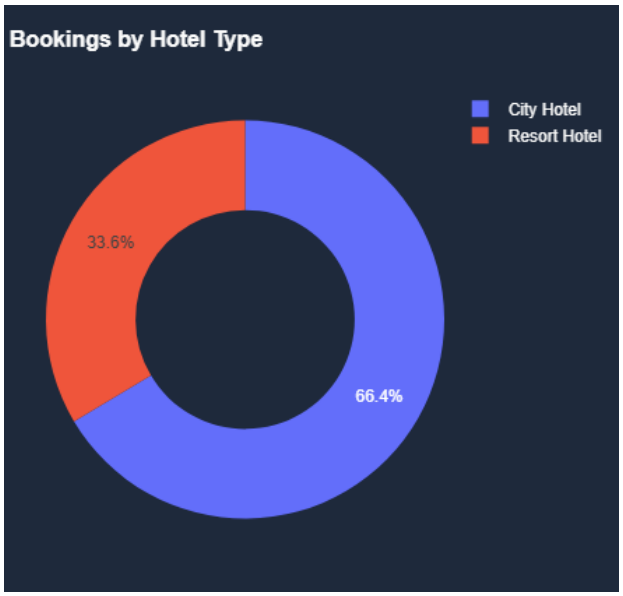


Figure 2: Booking Share by Hotel Type

11. Seasonality & Monthly Booking Trends

Seasonality plays an important role in hotel demand because booking volumes can vary considerably across different months of the year. Analysing monthly booking trends helps management understand when demand is strongest and when additional promotional or pricing strategies may be required. In this project, monthly booking activity was examined using the `arrival_date_month` variable to identify changes in demand throughout the year.

The analysis shows a clear variation in booking volume across the months. **October is identified as the peak booking month**, with approximately **13,877 bookings** in the supplied dashboard analysis. This indicates a period of relatively strong customer demand and provides an opportunity for hotels to improve revenue performance through appropriate pricing and inventory-management strategies.

The monthly trend also demonstrates that demand should not be treated as constant throughout the year. Lower-demand periods may require targeted marketing campaigns, promotional packages and customer incentives to improve occupancy. In contrast, high-demand periods require stronger inventory controls to avoid unnecessary discounts and to maximise revenue from available rooms.

Seasonality should also be considered alongside hotel type. City Hotel and Resort Hotel may experience different customer requirements and demand patterns because of differences in location, travel purpose and booking behaviour. Therefore, management should evaluate monthly performance at both the overall and property levels rather than relying only on total booking volume.

Monthly Demand Summary

Seasonality Indicator	Result	Business Interpretation
Peak Booking Month	**October**	Highest observed booking demand
October Bookings	**13,877**	Strong demand period
Analysis Period	**2017–2019**	Multi-year historical trend
Monthly Demand Pattern	**Variable**	Demand changes across the year

Overall, the monthly analysis confirms that **seasonality is an important factor in hotel demand planning**. The findings can support seasonal pricing, staffing, room-inventory planning and promotional strategies. Peak periods should be managed to maximise revenue, while weaker periods can be targeted with carefully designed demand-generation initiatives.

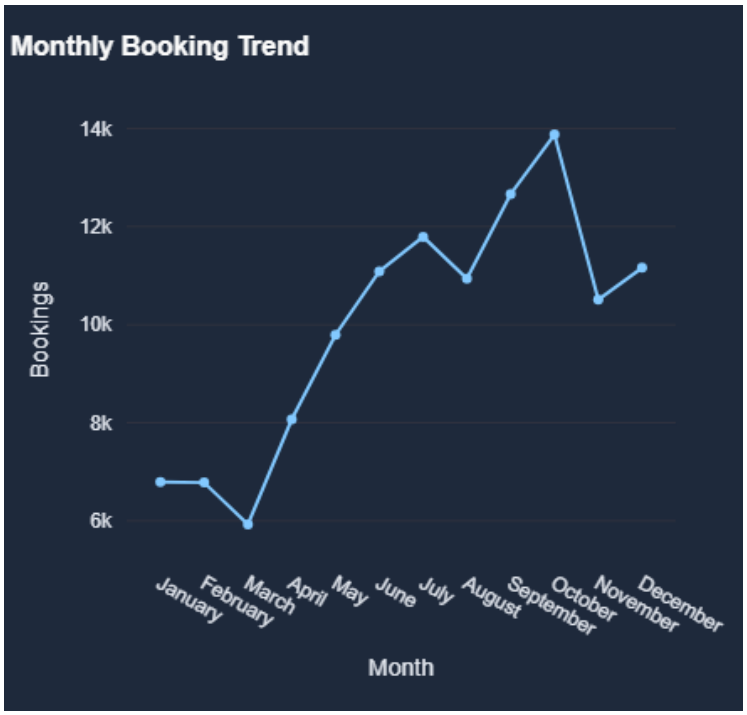


Figure 3: Monthly Booking Volume

12. Hotel Type Performance Analysis

The performance of the two hotel categories, **City Hotel and Resort Hotel**, was analysed to understand differences in booking demand and cancellation behaviour. Comparing the properties is important because each hotel type may attract different customer segments, travel purposes and booking patterns. A property-level comparison therefore provides more useful insights than analysing the combined dataset alone.

The analysis indicates that **City Hotel is the dominant booking category**, accounting for approximately **66.4% of total bookings**, while Resort Hotel contributes approximately **33.6%**. The higher booking volume suggests that City Hotel experiences stronger overall demand within the dataset. However, booking volume alone should not be considered a complete measure of performance because cancellations and other booking characteristics also influence effective occupancy and revenue.

Cancellation behaviour provides an additional performance dimension. The supplied dashboard analysis reports a cancellation rate of approximately **41.7% for City Hotel**, compared with approximately **27.8% for Resort Hotel**. This represents a substantial difference between the two properties and indicates that City Hotel carries greater cancellation exposure.

The difference may reflect variations in customer composition, booking channels, lead time, travel purpose or reservation policies. Therefore, management should avoid applying identical cancellation strategies to both hotel types. City Hotel may require stronger controls for high-risk reservations, while Resort Hotel can focus more heavily on seasonal demand management and occupancy optimisation.

Hotel Type Performance Summary

Performance Indicator	City Hotel	Resort Hotel	Business Interpretation
Booking Share	**66.4%**	**33.6%**	City Hotel has higher booking demand
Cancellation Rate	**41.7%**	**27.8%**	City Hotel has higher cancellation exposure
Relative Demand	Higher	Lower	Different capacity-planning requirements
Risk Priority	**High**	Moderate	Property-specific controls recommended

Overall, the comparison demonstrates that **City Hotel generates greater booking volume but also faces higher cancellation exposure**. This highlights the need for property-specific revenue and reservation-management strategies rather than a single approach across both hotel types.



Figure 4: Cancellation Rate by Hotel Type

13. Cancellation Behaviour Analysis

Cancellation behaviour is one of the most important areas of the hotel booking analysis because cancelled reservations can reduce expected occupancy, create inventory uncertainty and make revenue forecasting more difficult. The analysis therefore examines the overall cancellation rate and compares cancellation behaviour across hotel types and booking characteristics.

The supplied dashboard reports **119,390 total bookings**, of which approximately **37.0% were cancelled**. This indicates a significant level of cancellation exposure and highlights the importance of effective reservation and cancellation management. A high cancellation rate means that the number of reservations received cannot be treated as equivalent to confirmed occupancy demand.

Cancellation behaviour also differs between the two hotel categories. **City Hotel records a higher cancellation rate of approximately 41.7%**, compared with approximately **27.8% for Resort Hotel**. The difference suggests that cancellation risk is not evenly distributed across properties and that hotel-specific strategies may be more appropriate than a single cancellation policy.

Further analysis indicates that booking characteristics are also associated with cancellation behaviour. In particular, **booking lead time shows a strong relationship with cancellation risk**, with cancellations increasing substantially for reservations made further in advance. The supplied dashboard reports cancellation rates of approximately **9.6% for bookings made 0–7 days before arrival** and **57.0% for bookings made more than 180 days in advance**.

This pattern suggests that far-ahead bookings require closer monitoring and proactive confirmation. Hotels can reduce uncertainty by using targeted reminder messages, flexible rebooking options and appropriate cancellation or deposit policies for higher-risk reservations.

Cancellation Behaviour Summary

Cancellation Indicator	Result	Business Interpretation
Total Bookings	**119,390**	Overall reservation volume
Overall Cancellation Rate	**37.0%**	Significant cancellation exposure
City Hotel Cancellation Rate	**41.7%**	Higher property-level risk
Resort Hotel Cancellation Rate	**27.8%**	Lower cancellation exposure
0–7 Day Lead-Time Cancellation	**9.6%**	Lowest observed risk
181+ Day Lead-Time Cancellation	**57.0%**	Highest observed risk

Overall, the analysis demonstrates that **cancellation risk varies significantly across hotel types and booking lead-time groups**. These findings provide a strong basis for the subsequent Lead Time & Cancellation Risk analysis and targeted business recommendations.

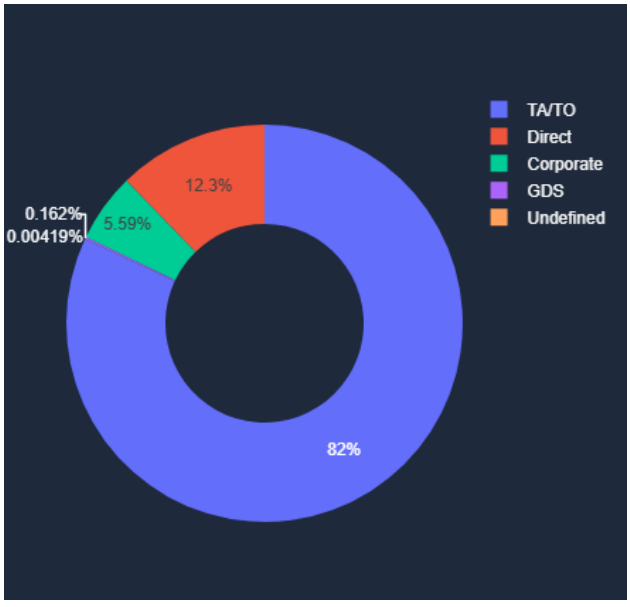


Figure 5: Cancellation Status Distribution

14·Stay Duration & Cancellation Analysis

Stay duration was analysed to determine whether the length of a planned hotel reservation is associated with different cancellation behaviour. This analysis uses the engineered **Total Stay Nights** variable, calculated by combining weekday and weekend nights. To make the comparison easier to interpret, bookings were grouped into **Short Stay (0–2 nights), Medium Stay (3–5 nights) and Long Stay (6+ nights)** categories.

The analysis indicates that cancellation behaviour varies across stay-duration groups. Short-stay reservations generally represent more straightforward booking requirements, while longer reservations involve a greater number of room nights and may therefore create greater exposure when cancelled. Comparing cancellation rates across the three categories helps identify whether longer reservations should receive additional monitoring.

Stay duration should also be interpreted together with hotel type. City Hotel and Resort Hotel serve different demand patterns, meaning that the relationship between length of stay and cancellation may not be identical across both properties. A segmented analysis therefore provides a more meaningful view than relying only on the overall cancellation rate.

The analysis is particularly useful from an operational perspective because a cancelled long-stay reservation can release a larger block of room inventory than a cancelled short-stay booking. If these cancellations occur close to the arrival date, replacing the lost room nights may be more difficult.

Stay Duration Analysis Summary

Stay Duration Category	Definition	Business Interpretation
Short Stay	0–2 nights	Lower room-night exposure per booking
Medium Stay	3–5 nights	Moderate booking commitment
Long Stay	6+ nights	Higher room-night exposure when cancelled
Total Stay Nights	Weekday + weekend nights	Core measure for stay-duration analysis

Overall, the stay-duration analysis provides an important additional dimension for understanding cancellation exposure. Rather than evaluating all cancelled bookings equally, management can consider both **cancellation probability and the number of room nights affected**. This supports more informed inventory planning and helps prioritise higher-impact reservations for monitoring and proactive communication.

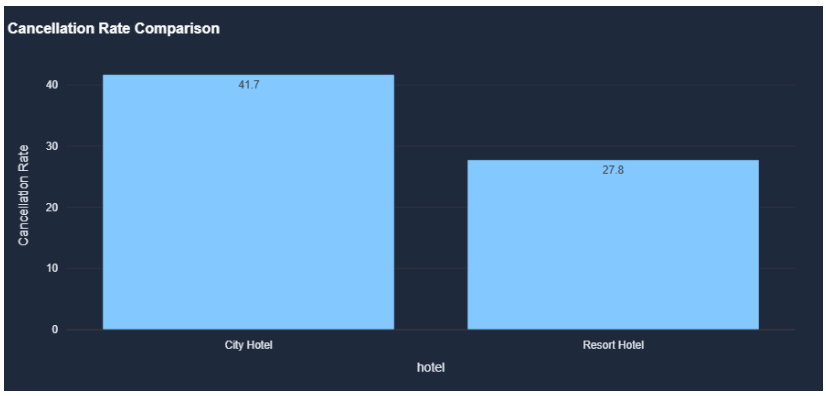


Figure 6: Cancellation Rate by Stay Category & Hotel Type

15·Booking Lead Time & Cancellation Risk

Booking lead time represents the number of days between the reservation date and the customer's scheduled arrival date. It is an important variable in hotel revenue management because it provides an indication of how far in advance customers make reservations. In this project, lead time was analysed to determine whether advance booking behaviour is associated with cancellation risk.

For business interpretation, bookings were grouped into five lead-time categories: **0–7 days, 8–30 days, 31–90 days, 91–180 days and 181+ days**. This segmentation makes it easier to compare cancellation behaviour across different booking windows and identify high-risk groups.

The analysis shows a clear relationship between **longer booking lead time and higher cancellation risk**. The supplied dashboard reports a cancellation rate of approximately **9.6% for bookings made 0–7 days before arrival**, while the rate increases to approximately **57.0% for bookings made more than 180 days in advance**. This represents a substantial increase in cancellation exposure as the booking window becomes longer.

The pattern has important implications for hotel management. Far-ahead bookings may provide early visibility of future demand, but they should not automatically be treated as guaranteed occupancy. A reservation made several months before arrival has more time for customer plans to change, increasing the probability of cancellation.

Hotels can use lead time as an early-warning indicator by applying additional monitoring to high-risk reservations. Appropriate actions may include confirmation reminders, flexible rebooking options, deposit requirements for selected segments and controlled overbooking strategies where operationally appropriate.

Lead-Time Risk Summary

Lead-Time Category	Cancellation Rate	Risk Interpretation
0–7 days	**9.6%**	Low cancellation risk
8–30 days	Increasing	Moderate risk
31–90 days	Increasing	Moderate-to-high risk
91–180 days	High	High cancellation exposure
181+ days	**57.0%**	Very high cancellation risk

Overall, **booking lead time is one of the strongest observable indicators of cancellation risk in the analysis**. Incorporating lead-time segments into reservation monitoring can help hotels improve occupancy predictability, reduce cancellation-related uncertainty and support more effective revenue-management decisions.

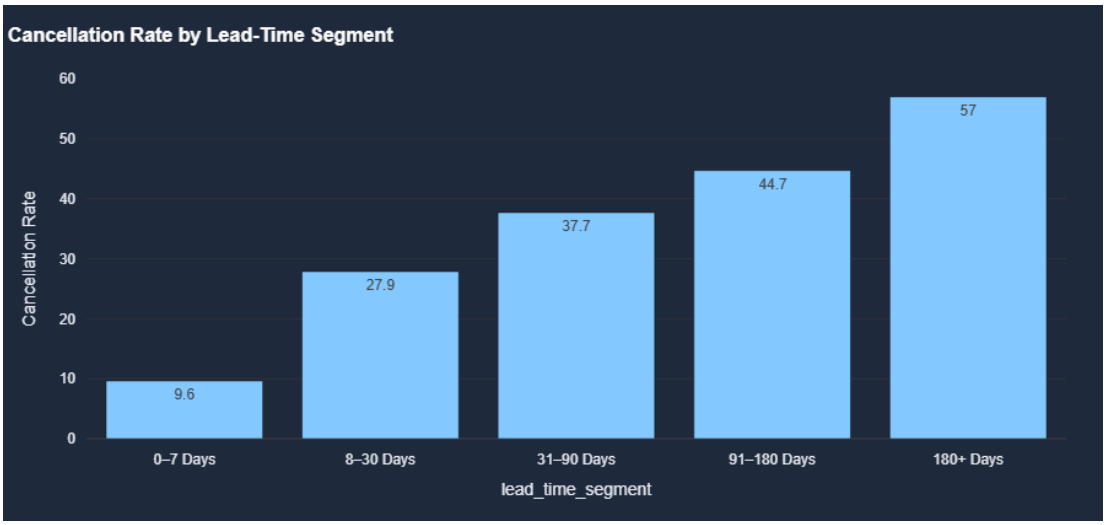


Figure 7: Cancellation Rate by Lead-Time Category

16. Deposit Type & Cancellation Analysis

Deposit type was analysed to understand whether the payment or security arrangement associated with a reservation is related to cancellation behaviour. The dataset contains three main deposit categories: **No Deposit, Non Refund and Refundable**. Examining these categories provides useful insight into how financial commitment at the time of booking may influence reservation stability.

The analysis shows that **No Deposit** is the dominant deposit arrangement in the dataset. This means that a large proportion of reservations are made without an upfront deposit requirement. From a business perspective, this provides customers with greater booking flexibility, but it may also create less financial commitment to the reservation and potentially increase exposure to cancellations.

The **Non Refund** category represents bookings where the reservation is subject to a non-refundable payment condition. Such bookings generally provide stronger financial commitment and can help reduce cancellation-related uncertainty. However, these policies may also influence customer booking decisions and should therefore be applied carefully across appropriate customer segments.

The **Refundable** category represents a comparatively small portion of bookings. Because of its limited volume, conclusions about cancellation behaviour for this category should be interpreted cautiously rather than treated as a major overall trend.

Deposit Type Analysis

Deposit Type	Booking Pattern	Business Interpretation
No Deposit	Dominant category	High booking flexibility; requires cancellation monitoring
Non Refund	Smaller segment	Stronger customer financial commitment
Refundable	Very limited segment	Flexible option; limited evidence for broad conclusions

The deposit analysis indicates that **payment commitment should be considered alongside lead time, hotel type and customer characteristics** when assessing cancellation risk. Rather than applying strict deposit requirements to all customers, hotels can use a targeted approach based on booking-risk characteristics. For example, higher-risk reservations may be candidates for stronger confirmation or deposit policies, while lower-risk bookings can retain greater flexibility.

Overall, deposit type provides an additional dimension for understanding cancellation exposure. However, it should not be interpreted as an independent cause of cancellation. The strongest business decisions should combine deposit information with **lead time, stay duration, hotel type and customer behaviour** to develop a more complete cancellation-risk strategy.

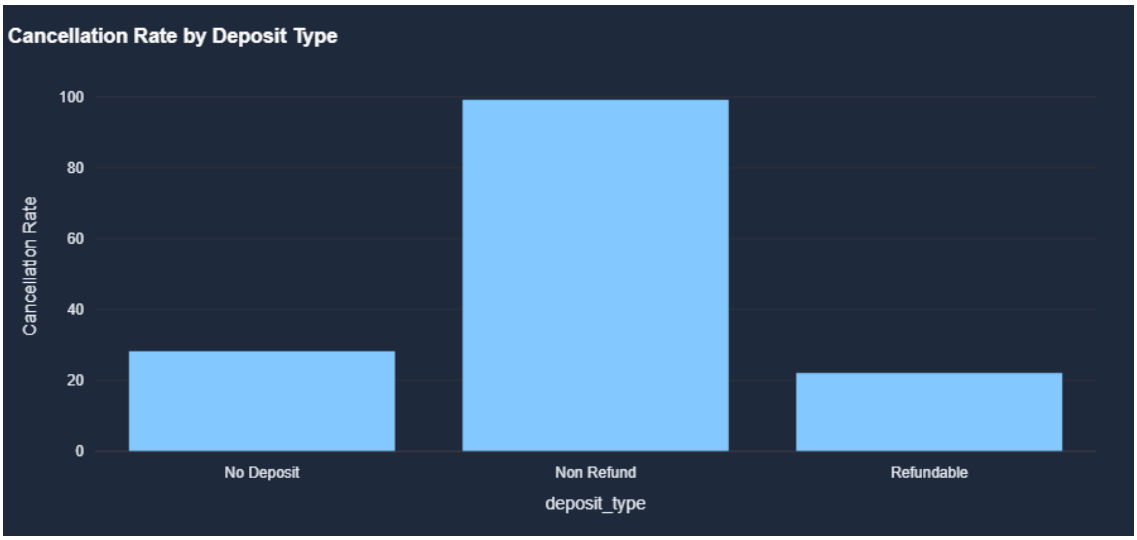


Figure 8: Cancellation Rate by Deposit Type

17. Market Segment & Booking Channel Analysis

Market segment and booking channel analysis was conducted to understand how customers make hotel reservations and which channels contribute to overall booking demand. This analysis is important because different booking channels may have different customer characteristics, lead times, cancellation behaviour and operational requirements.

The `market_segment` variable categorises reservations according to the source or commercial segment through which the booking was generated. The dataset includes segments such as **Online Travel Agents (OTA), Offline Travel Agents/Tour Operators, Direct, Groups, Corporate and Aviation**. Comparing these segments helps identify the major sources of hotel demand and provides a basis for evaluating channel performance.

The analysis indicates that **Online Travel Agents represent a major source of reservations** in the dataset. This highlights the importance of digital distribution channels in generating hotel demand. However, OTA bookings should be evaluated together with cancellation rates, lead time and customer acquisition considerations rather than only booking volume.

Direct and corporate bookings provide additional strategic value because they can represent opportunities for stronger customer relationships and potentially more controlled booking processes. Group and tour-operator reservations may also contribute significant demand but can have different operational requirements because of their booking size and planning horizon.

The analysis also considers `distribution_channel`, which provides another perspective on how reservations enter the hotel booking system. Comparing market segment and distribution channel allows management to distinguish between customer demand characteristics and the operational source of reservations.

Market Segment & Channel Summary

Analysis Dimension	Key Categories	Business Purpose
Market Segment	OTA, Direct, Corporate, Groups, Offline TA/TO, Aviation	Understand customer demand sources
Distribution Channel	TA/TO, Direct, Corporate, Undefined	Identify reservation channels
Booking Volume	Segment-wise booking count	Measure channel contribution
Cancellation	Segment/channel cancellation rate	Identify higher-risk channels
Lead Time	Segment-wise booking timing	Understand booking behaviour

Overall, market segment and booking channel analysis demonstrates that **booking source is an important dimension of hotel demand and cancellation management**. The findings can support channel optimisation, targeted marketing, customer acquisition planning and more effective management of high-risk reservation sources.

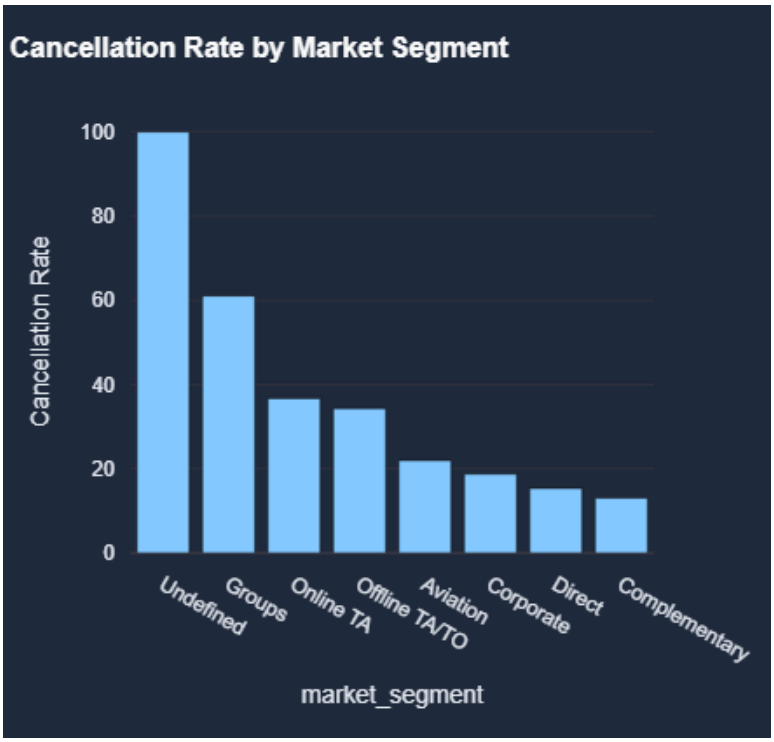


Figure 9: Cancellation Rate by Market Segment

18. Customer Behaviour & Repeat Guest Analysis

Customer behaviour analysis was conducted to understand the characteristics of guests making hotel reservations and to evaluate the contribution of repeat customers. The analysis uses variables such as **customer type**, `is_repeated_guest`, total guests and booking characteristics to identify differences in reservation behaviour.

The `customer_type` variable provides an important view of the purpose and nature of reservations. The dataset includes categories such as **Transient, Contract, Group and Transient-Party**. Among these, transient bookings represent individual or flexible travel arrangements, while contract and group bookings may involve different reservation requirements and planning horizons. Understanding these differences helps hotels design more appropriate pricing, communication and reservation strategies.

Repeat guest behaviour was examined using the `is_repeated_guest` indicator. Repeat customers are strategically important because they have already interacted with the hotel and may have greater familiarity with its services. From a business perspective, retaining existing customers can support more stable demand and create opportunities for personalised offers and loyalty initiatives.

The analysis also considers whether repeat-guest behaviour differs across hotel types and customer segments. Such comparisons can help identify customer groups that may be suitable for targeted retention programmes. However, repeat-guest status should be interpreted together with cancellation behaviour, booking channel and lead time rather than being treated as an independent measure of customer value.

Customer Behaviour Summary

Customer Behaviour Dimension	Key Variable / Category	Business Purpose
Customer Type	Transient, Contract, Group, Transient-Party	Understand booking characteristics
Repeat Guest	`is_repeated_guest`	Identify returning customers
Guest Composition	Adults, Children, Babies	Understand booking size
Hotel Preference	City Hotel / Resort Hotel	Compare customer demand
Booking Behaviour	Lead Time, Stay Duration	Understand reservation patterns
Cancellation Behaviour	`is_canceled`	Evaluate customer booking risk

Overall, the analysis highlights the importance of **understanding both new and returning customer behaviour**. Customer segmentation can help hotels develop targeted retention initiatives, personalised communication and more appropriate promotional strategies. Combining repeat-guest information with booking characteristics and cancellation risk can provide a stronger foundation for customer relationship and revenue-management decisions.

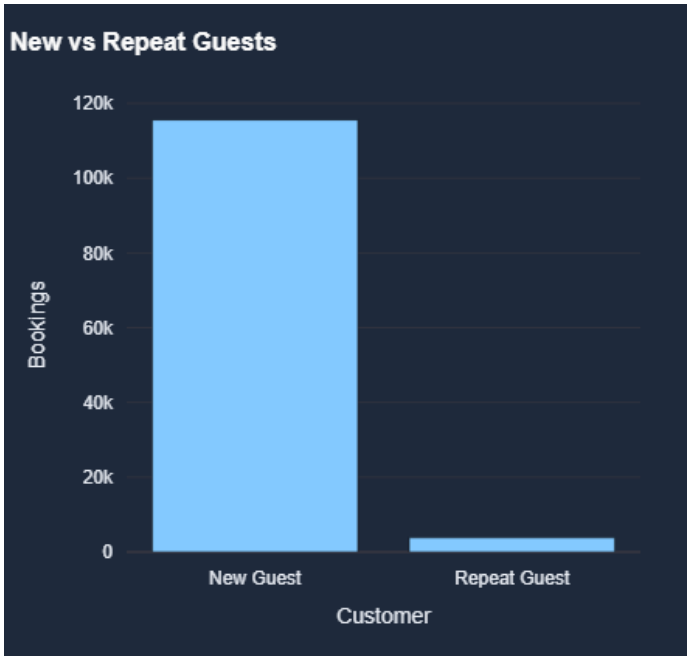


Figure 10- New vs Repeat Guest Cancellation

19·Key Business Insights

The analysis of hotel booking behaviour reveals several important patterns that have direct implications for hotel operations, revenue management and cancellation control. The findings demonstrate that booking volume alone is not sufficient to evaluate performance; hotels must also consider cancellation exposure, booking timing, seasonality and customer behaviour.

The first major insight is that **City Hotel is the dominant booking category**, accounting for approximately **66.4% of total bookings**. However, its higher demand is accompanied by a significantly higher cancellation rate of approximately **41.7%**, compared with **27.8% for Resort Hotel**. This indicates that City Hotel requires greater attention to reservation-risk management.

The second major insight is the strong relationship between **booking lead time and cancellation risk**. Cancellation increases from approximately **9.6% for bookings made 0–7 days before arrival to 57.0% for bookings made more than 180 days in advance**. Therefore, far-ahead reservations should not be treated as equally reliable occupancy commitments.

Seasonality is another important finding. **October is the peak booking month**, with approximately **13,877 bookings**, demonstrating that demand changes throughout the year. This creates opportunities for seasonal pricing, inventory optimisation and targeted promotions.

The analysis also shows that stay duration, deposit type, market segment and customer behaviour provide additional dimensions for understanding booking risk. These variables should be considered together rather than analysed independently.

Key Business Insights Summary

No.	Key Insight	Evidence	Business Implication
1	City Hotel dominates demand	**66.4% booking share**	Higher capacity and risk-management focus
2	City Hotel has higher cancellation exposure	**41.7% cancellation**	Property-specific controls required
3	Long lead time increases risk	**57.0% at 181+ days**	Monitor far-ahead bookings
4	Short lead-time bookings are lower risk	**9.6% at 0–7 days**	More reliable near-term demand
5	Demand is seasonal	**October: 13,877 bookings**	Strengthen seasonal planning
6	Booking characteristics influence risk	Stay, deposit, segment and customer factors	Use multi-factor risk assessment

Overall, the analysis indicates that **cancellation management should be proactive and data-driven**. Combining lead time, hotel type, stay duration, deposit arrangement and customer characteristics can help management identify higher-risk reservations and improve occupancy predictability.

20. Business Recommendations

Based on the analysis of hotel booking demand, seasonality, cancellation behaviour, lead time, stay duration, deposit type and customer characteristics, several practical recommendations can be proposed. The objective is to convert the analytical findings into actions that can improve occupancy predictability, reduce cancellation-related uncertainty and support better revenue management.

First, hotels should implement a **lead-time-based cancellation-risk management strategy**. Since cancellation risk increases substantially for bookings made far in advance, reservations in the **181+ day category** should receive additional monitoring and proactive confirmation. Automated reminders and flexible rebooking options can help reduce avoidable cancellations.

Second, **City Hotel should receive greater cancellation-management attention** because it has both the highest booking share and higher cancellation exposure than Resort Hotel. Management can develop property-specific reservation policies instead of applying identical controls across both hotel types.

Third, hotels should strengthen **seasonal demand planning**. With October identified as the peak booking month, room inventory, staffing and pricing should be planned ahead of high-demand periods. During weaker demand periods, targeted promotions and packages can be introduced to improve occupancy.

Fourth, management should consider **stay duration when prioritising cancellation risk**. Long-stay cancellations can affect a larger number of room nights, so these reservations may require additional monitoring.

Finally, booking source, deposit arrangement and customer behaviour should be incorporated into a broader **multi-factor risk framework**. Combining these variables can provide a more reliable basis for targeted communication, deposit policies and reservation management.

Recommendation Matrix

Recommendation	Evidence / Trigger	Priority	Expected Business Benefit
Lead-time risk monitoring	181+ day bookings show highest risk	**High**	Improve occupancy predictability
City Hotel controls	Higher booking volume and cancellation rate	**High**	Reduce cancellation exposure
Seasonal planning	October is peak demand period	**High**	Improve revenue and capacity utilisation
Long-stay monitoring	Greater room-night exposure	Medium	Protect inventory availability
Targeted deposit policies	Different deposit arrangements	Medium	Increase booking commitment
Channel-based management	Different market segments/channels	Medium	Improve channel efficiency
Customer retention initiatives	Repeat-guest behaviour	Medium	Strengthen customer loyalty

Overall, the recommendations support a shift from **reactive cancellation handling to proactive, data-driven reservation management**. Implementing these actions can help hotels balance customer flexibility with business protection while improving operational and revenue-planning decisions.

21. Expected Business Impact

The implementation of the recommendations developed from this analysis is expected to improve the way hotels manage reservations, cancellations, seasonal demand and room inventory. The primary business impact is expected to come from moving from a reactive approach to a **proactive, data-driven reservation management strategy**.

The analysis identified a significant overall cancellation exposure of approximately **37.0%**, with City Hotel recording a higher cancellation rate than Resort Hotel. By applying property-specific cancellation controls and monitoring high-risk reservations, management can improve the reliability of expected occupancy and reduce uncertainty in room-inventory planning.

The strongest opportunity comes from **booking lead-time management**. Since cancellation rates increase substantially for bookings made more than 180 days in advance, lead-time segmentation can be used as an early-warning mechanism. Proactive confirmation messages, targeted deposit policies and flexible rebooking options can help manage these reservations more effectively.

Seasonal demand planning can also improve resource utilisation. With **October identified as the peak booking month**, hotels can prepare room inventory, staffing and pricing strategies in advance. During lower-demand periods, targeted promotions can help stimulate demand without unnecessarily reducing prices during stronger periods.

The recommendations may also improve customer management by combining booking characteristics such as hotel type, stay duration, deposit arrangement, market segment and repeat-guest status. This enables management to prioritise high-risk or high-impact reservations rather than applying the same intervention to every customer.

Expected Business Impact Matrix

Impact Area	Expected Outcome	Business Value
Cancellation Management	Better identification of high-risk bookings	Lower uncertainty
Occupancy Planning	More reliable expected demand	Improved room utilisation
Revenue Management	Better seasonal pricing decisions	Revenue optimisation
Inventory Control	Earlier identification of booking risk	Reduced inventory disruption
Customer Management	Targeted communication and retention	Better customer experience
Operational Planning	Improved seasonal preparation	Efficient staffing and resources
Decision-Making	Data-driven reservation strategy	Stronger management decisions

Overall, the project is expected to provide a **structured analytical framework for improving hotel decision-making**. The actual financial impact would require implementation data and future-period measurement; therefore, the current analysis should be viewed as a basis for operational improvement rather than a quantified financial forecast.

22. Challenges & Limitations

The hotel booking analysis involved several data and analytical challenges that could influence the interpretation of results. Addressing these limitations was important to ensure that the findings were based on a structured and transparent analytical process.

One major challenge was **data quality**. The original dataset contained a substantial number of duplicate records and missing values across selected variables. Missing information in fields such as company, agent and city required careful treatment because simply deleting these observations could result in unnecessary data loss. Categorical inconsistencies, including undefined meal information, also required standardisation before visualisation and analysis.

Another challenge was the presence of **invalid or unusual observations**, including negative ADR values and bookings with zero recorded guests. These records required validation before being included in behavioural analysis because they could distort pricing and booking-related results.

The analysis also has limitations because it is based on **historical booking data from 2017–2019**. Customer behaviour, booking channels, pricing strategies and cancellation policies may change over time. Therefore, the findings should not automatically be assumed to represent current hotel-market conditions.

A further limitation is that the dataset primarily supports **descriptive and exploratory analysis**. The observed relationships between variables such as lead time and cancellation should not automatically be interpreted as causal relationships. For example, higher cancellation rates for long lead-time bookings do not prove that lead time alone causes cancellations.

Finally, the project does not include direct financial information such as actual hotel operating costs, profit margins or detailed revenue losses caused by cancellations. Therefore, the expected business impact is presented qualitatively rather than as a precise financial forecast.

Challenges & Limitations Summary

Challenge / Limitation	Impact on Analysis	Mitigation
Duplicate records	May inflate booking metrics	Duplicate removal
Missing values	Incomplete information	Context-based treatment
Invalid observations	May distort results	Validation and exclusion
Historical data	May not reflect current behaviour	Interpret findings within dataset period
Descriptive analysis	Cannot establish causation	Present relationships as associations
Limited financial data	Cannot calculate exact ROI	Provide qualitative business impact

Overall, these limitations do not invalidate the analysis but highlight the importance of **careful interpretation and future validation with current operational and financial data**.

23. Future Enhancements

The current hotel booking analysis provides a strong foundation for understanding booking demand, seasonality and cancellation behaviour. However, the project can be further enhanced by incorporating advanced analytical techniques, additional business data and more interactive decision-support capabilities. These enhancements would help transform the current descriptive analysis into a more predictive and strategic hotel business intelligence solution.

The first enhancement would be the development of a **cancellation prediction model** using machine learning. Variables such as lead time, hotel type, deposit type, market segment, stay duration, customer type and booking channel could be used to estimate the probability that an individual reservation will be cancelled. This would allow management to identify high-risk bookings before the arrival date.

A second enhancement would be the integration of **real-time or regularly updated booking data**. The current analysis is based on historical data from 2017–2019. Connecting the Streamlit dashboard to current reservation data would allow managers to monitor booking trends and cancellation risk continuously.

The project could also be extended with **revenue forecasting and dynamic pricing analysis**. Incorporating room availability, ADR, occupancy and revenue data would enable management to evaluate financial performance and optimise pricing during different demand periods.

Additional customer-level analysis could support **customer segmentation and retention strategies**, while integration with external factors such as holidays, events and weather could improve seasonal demand forecasting.

Future Enhancement Roadmap

Enhancement	Proposed Development	Expected Benefit
Cancellation Prediction	Machine learning risk model	Early identification of high-risk bookings
Real-Time Dashboard	Connect live/updated booking data	Continuous performance monitoring
Revenue Forecasting	Forecast occupancy, ADR and revenue	Better financial planning
Dynamic Pricing	Demand-based pricing analysis	Revenue optimisation
Customer Segmentation	Cluster customers by behaviour	Targeted marketing
External Data Integration	Holidays, events and weather	Improved demand forecasting
Advanced Alerts	Automated high-risk booking alerts	Faster management action
Dashboard Expansion	More interactive KPIs and filters	Better decision support

Overall, these enhancements can evolve the project from a **historical descriptive analytics solution into a predictive hotel business intelligence platform**, enabling management to make faster, more targeted and data-driven decisions.

24. Tools & Technologies

The Hotel Booking Analytics project was developed using a combination of **Python-based data analytics, statistical analysis, data visualisation and interactive dashboard technologies**. Each tool was selected according to its role in the data-analysis workflow, from data preparation and exploratory analysis to business visualisation and dashboard deployment.

Python was used as the primary programming language for data analysis and project development. It provided the foundation for data manipulation, preprocessing, feature engineering and analytical calculations.

Pandas was used extensively for loading the hotel booking dataset, inspecting data quality, handling missing values, removing duplicate records, transforming variables and creating analytical features. It supported the overall data-cleaning and preprocessing workflow.

NumPy supported numerical operations and calculations required during data preparation and analysis. It was particularly useful for handling numerical values and performing efficient array-based operations.

Matplotlib and **Seaborn** were used to create analytical visualisations during exploratory data analysis. Matplotlib provided flexible plotting capabilities, while Seaborn supported statistical and distribution-based visualisations that helped identify patterns and relationships in the dataset.

Streamlit was used to transform the analytical work into an interactive business dashboard. The dashboard presents key performance indicators, booking trends, cancellation analysis and business insights in a user-friendly format, allowing users to explore the results interactively.

Technology Stack

Tool / Technology	Primary Role	Project Application
Python	Programming & Analysis	Core analytical workflow
Pandas	Data Manipulation	Cleaning, transformation & feature engineering
NumPy	Numerical Computing	Numerical calculations
Matplotlib	Visualisation	EDA charts and analytical plots
Seaborn	Statistical Visualisation	Distribution and relationship analysis
Streamlit	Dashboard Development	Interactive hotel analytics dashboard
Google Colab	Development Environment	Notebook-based analysis and experimentation

Together, these technologies provide an end-to-end analytical workflow covering **data preparation, exploratory analysis, visualisation, business interpretation and interactive reporting**. The combination also demonstrates practical skills in Python-based data analytics and business intelligence development.

25. Conclusion

The **Hotel Booking & Cancellation Behaviour Analysis** project demonstrates how historical hotel reservation data can be transformed into meaningful business insights using a structured data analytics approach. The analysis covered **119,390 hotel booking records from 2017–2019**, focusing on booking demand, seasonality, hotel type, stay duration, lead time, deposit type, market segment and customer behaviour.

The findings show that **City Hotel accounts for the larger share of bookings at approximately 66.4%**, but it also records a higher cancellation rate of approximately **41.7%**, compared with **27.8% for Resort Hotel**. This highlights the importance of evaluating booking volume together with cancellation exposure when assessing hotel performance.

One of the most significant findings is the relationship between **booking lead time and cancellation risk**. Cancellation rates increase considerably as reservations are made further in advance, reaching approximately **57.0% for bookings with a lead time of more than 180 days**, compared with approximately **9.6% for bookings made 0–7 days before arrival**. This provides a useful basis for proactive reservation-risk management.

The analysis also identified seasonal demand variation, with **October recording the highest booking volume** in the project analysis. These patterns can support improved inventory planning, seasonal pricing and promotional strategies.

Overall, the project demonstrates the value of **Python, Pandas, NumPy, Matplotlib, Seaborn and Streamlit** in developing an end-to-end analytics solution. The resulting insights provide a foundation for more proactive cancellation management, demand planning and data-driven hotel decision-making. Future integration of machine learning, real-time data and revenue forecasting could further strengthen the solution.

26. References

The following sources and documentation were referred to during the development of the **Hotel Booking & Cancellation Behaviour Analysis** project. These references supported the data analysis, preprocessing, visualisation, dashboard development and interpretation of the hotel booking dataset.

1. **Hotel Booking Demand Dataset**
   Antonio, N., Almeida, A., & Nunes, L. (2019). *Hotel Booking Demand Dataset*. Data in Brief, 22, 41–49. The dataset was used as the primary source for analysing hotel booking patterns, cancellations, customer behaviour and seasonal demand.
2. **Pandas Documentation**
   Official Pandas documentation was referred to for data loading, DataFrame operations, missing-value treatment, duplicate removal, filtering, grouping and data transformation.
3. **NumPy Documentation**
   NumPy documentation was used as a reference for numerical operations and array-based calculations during the data-preprocessing and analytical stages.
4. **Matplotlib Documentation**
   Matplotlib documentation supported the development and customisation of analytical charts used during exploratory data analysis.
5. **Seaborn Documentation**
   Seaborn documentation was consulted for statistical visualisations, distribution analysis and relationship-based plots.
6. **Streamlit Documentation**
   Official Streamlit documentation was used for developing the interactive hotel analytics dashboard and implementing dashboard components, layouts and visual elements.
7. **Google Colab**
   Google Colaboratory was used as the notebook-based development environment for data preparation, exploratory analysis, visualisation and experimentation.

These sources provided the technical and dataset foundations required to complete the project using a structured **Python-based data analytics workflow**.

27. Appendix

The appendix contains supporting materials used during the development and analysis of the **Hotel Booking & Cancellation Behaviour Analysis** project. These materials provide additional technical and analytical details that support the findings presented in the main report while keeping the primary sections focused on business insights and conclusions.

The project appendix includes selected outputs from the **Google Colab analysis**, data-cleaning procedures, feature-engineering calculations, exploratory visualisations and supporting statistical summaries. It may also include screenshots or selected views of the **Streamlit interactive dashboard** developed for presenting key performance indicators, booking trends and cancellation analysis.

Key supporting materials include:

- **Appendix A:** Dataset structure and column information.
- **Appendix B:** Data-quality and missing-value assessment.
- **Appendix C:** Data-cleaning and preprocessing outputs.
- **Appendix D:** Feature-engineering calculations and derived variables.
- **Appendix E:** Exploratory Data Analysis visualisations.
- **Appendix F:** Hotel type and seasonal booking analysis.
- **Appendix G:** Cancellation and booking lead-time analysis.
- **Appendix H:** Supporting customer, deposit and market-segment analysis.
- **Appendix I:** Streamlit dashboard screenshots and KPI outputs.
- **Appendix J:** Selected Python/Google Colab code used for analysis.

The appendix serves as a **technical reference section** for readers who wish to examine the analytical process in greater detail. The complete working notebook and dashboard implementation can be maintained separately as project deliverables.